KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5283A | 5 Credits: 4 | Program Elective | Source-grounded

Data and Application Security

● Understand symmetric and asymmetric cryptographic systems and cryptology through

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5283A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5283A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Cyber Forensics And Information security
Course code	5283A
Course title	Data and Application Security
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- ● Understand symmetric and asymmetric cryptographic systems and cryptology through
- cryptography techniques and algorithms.
- ● Explain the hash function as an application of cryptography with the concept of
- message integrity and digital signature authentication.
- ● Understand security concepts, security professional roles, and security resources in
- the context of systems and security development life cycle.

Course outcomes

CO	Official outcome	Study level
CO1	systems and cryptology through 14 Applying cryptography techniques and algorithms. Understand asymmetric cryptographic	See official syllabus
CO2	systems and cryptology through 18 Applying cryptography techniques and algorithms. Explain the hash function as an application of cryptography with the concept of message	See official syllabus
CO3	16 Applying integrity and digital signature authentication. Describe various security threats to web	See official syllabus
CO4	applications/ servers and providing security to 10 Understanding web servers.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 2
CO2	CO2 3 2
CO3	CO3 3 2

Row	Extracted official mapping
CO4	CO4 3 1

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	cryptography techniques and algorithms.	3	As specified
Module 2	cryptography techniques and algorithms.	3	As specified
Module 3	concept of message integrity and digital signature authentication.	4	As specified
Module 4	providing security to web servers.	3	As specified

MODULE 1

cryptography techniques and algorithms.

This module is presented from the official syllabus scope for Data and Application Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: cryptography techniques and algorithms.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain basics of cryptography 2 Understanding Illustrate Symmetric Key Cryptographic

Explain basics of cryptography 2 Understanding Illustrate Symmetric Key Cryptographic is an official topic in Module 1 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain basics of cryptography 2 Understanding Illustrate Symmetric Key Cryptographic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain basics of cryptography 2 understanding illustrate symmetric key cryptographic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain basics of cryptography 2 Understanding Illustrate Symmetric Key Cryptographic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain basics of cryptography 2 Understanding Illustrate Symmetric Key Cryptographic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

5 Understanding Systems Solve problems of Symmetric key

5 Understanding Systems Solve problems of Symmetric key is an official topic in Module 1 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding Systems Solve problems of Symmetric key using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding systems solve problems of symmetric key should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding Systems Solve problems of Symmetric key means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ 5 Understanding Systems Solve problems of Symmetric key □□□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

7 Applying Cryptographic Systems Probability theory -Information theory – Complexity theory, Number theory. Symmetric (Private) Key Cryptographic Systems: Caesar – Affine – Monoalphabetic Substitution – Transposition – Homophonic substitution – Vignere – Beaufort and DES Family – Product ciphers – Lucifer and DES. Understand asymmetric cryptographic systems and cryptology through

7 Applying Cryptographic Systems Probability theory -Information theory – Complexity theory, Number theory. Symmetric (Private) Key Cryptographic Systems: Caesar – Affine – Monoalphabetic Substitution – Transposition – Homophonic substitution – Vignere – Beaufort and DES Family – Product ciphers – Lucifer and DES. Understand asymmetric cryptographic systems and cryptology through is an official topic in Module 1 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 7 Applying Cryptographic Systems Probability theory -Information theory – Complexity theory, Number theory. Symmetric (Private) Key Cryptographic Systems: Caesar – Affine – Monoalphabetic Substitution – Transposition – Homophonic substitution – Vignere – Beaufort and DES Family – Product ciphers – Lucifer and DES. Understand asymmetric cryptographic systems and cryptology through using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 7 applying cryptographic systems probability theory -information theory – complexity theory, number theory. symmetric (private) key cryptographic systems: caesar – affine – monoalphabetic substitution – transposition – homophonic substitution – vignere – beaufort and des family – product ciphers – lucifer and des. understand asymmetric cryptographic systems and cryptology through should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 7 Applying Cryptographic Systems Probability theory -Information theory – Complexity theory, Number theory. Symmetric (Private) Key Cryptographic Systems: Caesar – Affine – Monoalphabetic Substitution – Transposition – Homophonic substitution – Vignere – Beaufort and DES Family – Product ciphers – Lucifer and DES. Understand asymmetric cryptographic systems and cryptology through means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 7 Applying Cryptographic Systems Probability theory -Information theory – Complexity theory, Number theory. Symmetric (Private) Key Cryptographic Systems: Caesar – Affine – Monoalphabetic Substitution – Transposition – Homophonic substitution – Vignere – Beaufort and DES Family – Product ciphers – Lucifer and DES. Understand asymmetric cryptographic systems and cryptology through [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
cryptography techniques and algorithms.
 M1.01      Explain basics of cryptography                           2
Understanding
            Illustrate   Symmetric     Key     Cryptographic
 M1.02                                                               5
Understanding
            Systems
            Solve    problems     of       Symmetric        key
 M1.03                                                               7
Applying
            Cryptographic Systems
Contents: Introduction: History of Cryptography. Mathematical background:
Probability theory -Information theory – Complexity theory, Number theory.
Symmetric
(Private) Key Cryptographic Systems: Caesar – Affine – Monoalphabetic Substitution
–
Transposition – Homophonic substitution – Vignere – Beaufort and DES Family –
Product ciphers – Lucifer and DES.
            Understand asymmetric cryptographic systems and cryptology through
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

cryptography techniques and algorithms.

This module is presented from the official syllabus scope for Data and Application Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: cryptography techniques and algorithms.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

4 Understanding Systems Solve problems of Asymmetric key

4 Understanding Systems Solve problems of Asymmetric key is an official topic in Module 2 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding Systems Solve problems of Asymmetric key using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding systems solve problems of asymmetric key should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding Systems Solve problems of Asymmetric key means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ 4 Understanding Systems Solve problems of Asymmetric key □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□□.

8 Applying Cryptographic Systems Explain the concepts of Stream and block

8 Applying Cryptographic Systems Explain the concepts of Stream and block is an official topic in Module 2 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 8 Applying Cryptographic Systems Explain the concepts of Stream and block using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 8 applying cryptographic systems explain the concepts of stream and block should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 8 Applying Cryptographic Systems Explain the concepts of Stream and block means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ 8 Applying Cryptographic Systems Explain the concepts of Stream and block □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

6 Understanding ciphers Contents :Asymmetric (Public) Key Cryptographic Systems: Concept of PKCS, RSA Cryptosystem- Variants of RSA – Primality testing – Security of RSA – Merkle – Hellman – Security of Merkle – Hellman, ElGamal. Elliptical Curve Cryptography. Stream ciphers and block ciphers: The one time pad – Synchronous stream ciphers – Self-synchronizing stream ciphers – Feedback shift registers – Linear Complexity – Non-linear feedback shift registers – Stream ciphers based LFSRs. Explain the hash function as an application of cryptography with the

6 Understanding ciphers Contents :Asymmetric (Public) Key Cryptographic Systems: Concept of PKCS, RSA Cryptosystem- Variants of RSA – Primality testing – Security of RSA – Merkle – Hellman – Security of Merkle – Hellman, ElGamal. Elliptical Curve Cryptography. Stream ciphers and block ciphers: The one time pad – Synchronous stream ciphers – Self-synchronizing stream ciphers – Feedback shift registers – Linear Complexity – Non-linear feedback shift registers – Stream ciphers based LFSRs. Explain the hash function as an application of cryptography with the is an official topic in Module 2 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding ciphers Contents :Asymmetric (Public) Key Cryptographic Systems: Concept of PKCS, RSA Cryptosystem- Variants of RSA – Primality testing – Security of RSA – Merkle – Hellman – Security of Merkle – Hellman, ElGamal. Elliptical Curve Cryptography. Stream ciphers and block ciphers: The one time pad – Synchronous stream ciphers – Self-synchronizing stream ciphers – Feedback shift registers – Linear Complexity – Non-linear feedback shift registers – Stream ciphers based LFSRs. Explain the hash function as an application of cryptography with the using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding ciphers contents :asymmetric (public) key cryptographic systems: concept of pkcs, rsa cryptosystem- variants of rsa – primality testing – security of rsa – merkle – hellman – security of merkle – hellman, elgamal. elliptical curve cryptography. stream ciphers and block ciphers: the one time pad – synchronous stream ciphers – self-synchronizing stream ciphers – feedback shift registers – linear complexity – non-linear feedback shift registers – stream ciphers based lfsrs. explain the hash function as an application of cryptography with the should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding ciphers Contents :Asymmetric (Public) Key Cryptographic Systems: Concept of PKCS, RSA Cryptosystem- Variants of RSA – Primality testing – Security of RSA – Merkle – Hellman – Security of Merkle – Hellman, ElGamal. Elliptical Curve Cryptography. Stream ciphers and block ciphers: The one time pad – Synchronous stream ciphers – Self-synchronizing stream ciphers – Feedback shift registers – Linear Complexity – Non-linear feedback shift registers – Stream ciphers based LFSRs. Explain the hash function as an application of cryptography with the means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 6 Understanding ciphers Contents :Asymmetric (Public) Key Cryptographic Systems: Concept of PKCS, RSA Cryptosystem- Variants of RSA – Primality testing – Security of RSA – Merkle – Hellman – Security of Merkle – Hellman, ElGamal. Elliptical Curve Cryptography. Stream ciphers and block ciphers: The one time pad – Synchronous stream ciphers – Self-synchronizing stream ciphers – Feedback shift registers – Linear Complexity – Non-linear feedback shift registers – Stream ciphers based LFSRs. Explain the hash function as an application of cryptography with the □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
cryptography techniques and algorithms.
            Illustrate   Asymmetric     Key    Cryptographic
 M2.01                                                              4
Understanding
            Systems
            Solve   problems    of        Asymmetric        key
 M2.02                                                              8
Applying
            Cryptographic Systems
            Explain the concepts of Stream and block
 M2.03                                                              6
Understanding
            ciphers

            Series Test – I                                         1
Contents :Asymmetric (Public) Key Cryptographic Systems: Concept of PKCS, RSA
Cryptosystem- Variants of RSA – Primality testing – Security of RSA – Merkle –
Hellman – Security of Merkle – Hellman, ElGamal. Elliptical Curve Cryptography.
Stream ciphers and block ciphers: The one time pad – Synchronous stream ciphers –
Self-synchronizing stream ciphers – Feedback shift registers – Linear Complexity –
Non-linear feedback shift registers – Stream ciphers based LFSRs.
            Explain the hash function as an application of cryptography with the
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

concept of message integrity and digital signature authentication.

This module is presented from the official syllabus scope for Data and Application Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours | see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: concept of message integrity and digital signature authentication.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Illustrate the concepts of Digital signatures

Illustrate the concepts of Digital signatures is an official topic in Module 3 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the concepts of Digital signatures using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the concepts of digital signatures should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the concepts of Digital signatures means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Illustrate the concepts of Digital signatures [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Explain various secret sharing algorithms

Explain various secret sharing algorithms is an official topic in Module 3 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain various secret sharing algorithms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain various secret sharing algorithms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain various secret sharing algorithms means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Explain various secret sharing algorithms [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate random number generators 3 Understanding Solve problems of digital signatures, secret

Illustrate random number generators 3 Understanding Solve problems of digital signatures, secret is an official topic in Module 3 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate random number generators 3 Understanding Solve problems of digital signatures, secret using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate random number generators 3 understanding solve problems of digital signatures, secret should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate random number generators 3 Understanding Solve problems of digital signatures, secret means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Illustrate random number generators 3 Understanding Solve problems of digital signatures, secret □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

sharing algorithms and random number 5 Applying generators Contents : Digital Signatures: Properties, Generic signature schemes – Rabin - Lamport – Matyas Meyer, RSA – Multiple RSA and ElGamal Signatures – Digital signature standard – Blind Signatures- RSA Blind. Secret Sharing Algorithms: Threshold secret sharing – Shamir scheme, Blakley scheme and modular Scheme. Pseudo random number generators: Definition of randomness and pseudo-randomness – Statistical tests of randomness – Linear congruential generator – Modern PRNGs (a brief description). Describe various security threats to web applications/ servers and

sharing algorithms and random number 5 Applying generators Contents : Digital Signatures: Properties, Generic signature schemes – Rabin - Lamport – Matyas Meyer, RSA – Multiple RSA and ElGamal Signatures – Digital signature standard – Blind Signatures- RSA Blind. Secret Sharing Algorithms: Threshold secret sharing – Shamir scheme, Blakley scheme and modular Scheme. Pseudo random number generators: Definition of randomness and pseudo-randomness – Statistical tests of randomness – Linear congruential generator – Modern PRNGs (a brief description). Describe various security threats to web applications/ servers and is an official topic in Module 3 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify sharing algorithms and random number 5 Applying generators Contents : Digital Signatures: Properties, Generic signature schemes – Rabin - Lamport – Matyas Meyer, RSA – Multiple RSA and ElGamal Signatures – Digital signature standard – Blind Signatures- RSA Blind. Secret Sharing Algorithms: Threshold secret sharing – Shamir scheme, Blakley scheme and modular Scheme. Pseudo random number generators: Definition of randomness and pseudo-randomness – Statistical tests of randomness – Linear congruential generator – Modern PRNGs (a brief description). Describe various security threats to web applications/ servers and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, sharing algorithms and random number 5 applying generators contents : digital signatures: properties, generic signature schemes – rabin - lamport – matyas meyer, rsa – multiple rsa and elgamal signatures – digital signature standard – blind signatures- rsa blind. secret sharing algorithms: threshold secret sharing – shamir scheme, blakley scheme and modular scheme. pseudo random number generators: definition of randomness and pseudo-randomness – statistical tests of randomness – linear congruential generator – modern prngs (a brief description). describe various security threats to web applications/ servers and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What sharing algorithms and random number 5 Applying generators Contents : Digital Signatures: Properties, Generic signature schemes – Rabin - Lamport – Matyas Meyer, RSA – Multiple RSA and ElGamal Signatures – Digital signature standard – Blind Signatures- RSA Blind. Secret Sharing Algorithms: Threshold secret sharing – Shamir scheme, Blakley scheme and modular Scheme. Pseudo random number generators: Definition of randomness and pseudo-randomness – Statistical tests of randomness – Linear congruential generator – Modern PRNGs (a brief description). Describe various security threats to web applications/ servers and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ sharing algorithms and random number 5 Applying generators Contents : Digital Signatures: Properties, Generic signature schemes – Rabin - Lamport – Matyas Meyer, RSA – Multiple RSA and ElGamal Signatures – Digital signature standard – Blind Signatures- RSA Blind. Secret Sharing Algorithms: Threshold secret sharing – Shamir scheme, Blakley scheme and modular Scheme. Pseudo random number generators: Definition of randomness and pseudo-randomness – Statistical tests of randomness – Linear congruential generator – Modern PRNGs (a brief description). Describe various security threats to web applications/ servers and □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
concept of message integrity and digital signature authentication.

 M3.01      Illustrate the concepts of Digital signatures            4
Understanding
  M3.02     Explain various secret sharing algorithms             4
Understanding

  M3.03     Illustrate random number generators                   3
Understanding
               Solve problems of digital signatures, secret
   M3.04       sharing algorithms and random number                  5
Applying
               generators
 Contents : Digital Signatures: Properties, Generic signature schemes – Rabin -
 Lamport – Matyas Meyer, RSA – Multiple RSA and ElGamal Signatures – Digital
 signature standard – Blind Signatures- RSA Blind. Secret Sharing Algorithms:
 Threshold secret sharing – Shamir scheme, Blakley scheme and modular Scheme.
 Pseudo random number generators: Definition of randomness and pseudo-randomness
 – Statistical tests of randomness – Linear congruential generator – Modern PRNGs
(a
 brief description).
            Describe various security threats to web applications/ servers and
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

providing security to web servers.

This module is presented from the official syllabus scope for Data and Application Security. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: providing security to web servers.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

4 Understanding defense mechanisms

4 Understanding defense mechanisms is an official topic in Module 4 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding defense mechanisms using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding defense mechanisms should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding defense mechanisms means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] 4 Understanding defense mechanisms [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Illustrate the mechanism of transmitting data 2 Understanding Explain vulnerabilities of access control and

Illustrate the mechanism of transmitting data 2 Understanding Explain vulnerabilities of access control and is an official topic in Module 4 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the mechanism of transmitting data 2 Understanding Explain vulnerabilities of access control and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the mechanism of transmitting data 2 understanding explain vulnerabilities of access control and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the mechanism of transmitting data 2 Understanding Explain vulnerabilities of access control and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Illustrate the mechanism of transmitting data 2 Understanding Explain vulnerabilities of access control and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding securing mechanisms. Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls Text / Reference T/R Book Title/Author Cryptography and Network Security: Principles and Practice, Global Edition, T1 7/E, William Stallings, Pearson Padmanabhan T R, Shyamala C and Harini N, "Cryptography and Security", T2 Wiley Publications 2011. DafyddStuttard, Marcus Pinto, The Web Application Hacker's Handbook, 2nd T3 Edition, Wiley Publishing, Inc. Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com

4 Understanding securing mechanisms. Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls Text / Reference T/R Book Title/Author Cryptography and Network Security: Principles and Practice, Global Edition, T1 7/E, William Stallings, Pearson Padmanabhan T R, Shyamala C and Harini N, "Cryptography and Security", T2 Wiley Publications 2011. DafyddStuttard, Marcus Pinto, The Web Application Hacker's Handbook, 2nd T3 Edition, Wiley Publishing, Inc. Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com is an official topic in Module 4 of Data and Application Security. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding securing mechanisms. Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls Text / Reference T/R Book Title/Author Cryptography and Network Security: Principles and Practice, Global Edition, T1 7/E, William Stallings, Pearson Padmanabhan T R, Shyamala C and Harini N, “Cryptography and Security”, T2 Wiley Publications 2011. DafyddStuttard, Marcus Pinto, The Web Application Hacker’s Handbook, 2nd T3 Edition, Wiley Publishing, Inc. Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding securing mechanisms. contents :web application security- key problem factors – core defense mechanisms- handling user access- handling user input- handling attackers – web spidering – discovering hidden content. transmitting data via the client – hidden form fields – http cookies – url parameters – handling client-side data securely – attacking authentication – design flaws in authentication mechanisms –securing authentication attacking access controls – common vulnerabilities – securing access controls text / reference t/r book title/author cryptography and network security: principles and practice, global edition, t1 7/e, william stallings, pearson padmanabhan t r, shyamala c and harini n, “cryptography and security”, t2 wiley publications 2011. dafyddstuttard, marcus pinto, the web application hacker’s handbook, 2nd t3 edition, wiley publishing, inc. online resources sl.no website link 1 https://www.tutorialspoint.com should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding securing mechanisms. Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls Text / Reference T/R Book Title/Author Cryptography and Network Security: Principles and Practice, Global Edition, T1 7/E, William Stallings, Pearson Padmanabhan T R, Shyamala C and Harini N, “Cryptography and Security”, T2 Wiley Publications 2011. DafyddStuttard, Marcus Pinto, The Web Application Hacker’s Handbook, 2nd T3 Edition, Wiley Publishing, Inc. Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 4 Understanding securing mechanisms. Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms –securing authentication Attacking access controls – Common vulnerabilities – Securing access controls Text / Reference T/R Book Title/Author Cryptography and Network Security: Principles and Practice, Global Edition, T1 7/E, William Stallings, Pearson Padmanabhan T R, Shyamala C and Harini N, “Cryptography and Security”, T2 Wiley Publications 2011. DafyddStuttard, Marcus Pinto, The Web Application Hacker’s Handbook, 2nd T3 Edition, Wiley Publishing, Inc. Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
providing security to web servers.
            Explain web application security threats and
M4.01                                                       4
Understanding
            defense mechanisms
M4.02       Illustrate the mechanism of transmitting data         2
Understanding
            Explain vulnerabilities of access control and
M4.03                                                       4
Understanding
            securing mechanisms.
             Series Test – II                               1
Contents :Web application security- Key Problem factors – Core defense
mechanisms-
Handling user access- handling user input- Handling attackers – web spidering –
Discovering hidden content. Transmitting data via the client – Hidden form fields
–
HTTP cookies – URL parameters – Handling client-side data securely – Attacking
authentication – design flaws in authentication mechanisms –securing
authentication
Attacking access controls – Common vulnerabilities – Securing access controls

Text / Reference


T/R                                    Book Title/Author
        Cryptography and Network Security: Principles and Practice, Global
Edition,
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain basics of cryptography 2 Understanding Illustrate Symmetric Key Cryptographic. (3 marks)

Answer: Begin with the standard definition or identity of *Explain basics of cryptography 2 Understanding Illustrate Symmetric Key Cryptographic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 5 Understanding Systems Solve problems of Symmetric key. (3 marks)

Answer: Begin with the standard definition or identity of *5 Understanding Systems Solve problems of Symmetric key*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 7 Applying Cryptographic Systems Probability theory -Information theory – Complexity theory, Number theory. Symmetric (Private) Key Cryptographic Systems: Caesar – Affine – Monoalphabetic Substitution – Transposition – Homophonic substitution – Vignere – Beaufort and DES Family – Product ciphers – Lucifer and DES. Understand asymmetric cryptographic systems and cryptology through. (3 marks)

Answer: Begin with the standard definition or identity of *7 Applying Cryptographic Systems Probability theory -Information theory – Complexity theory, Number theory. Symmetric (Private) Key Cryptographic Systems: Caesar – Affine – Monoalphabetic Substitution – Transposition – Homophonic substitution – Vignere – Beaufort and DES Family – Product ciphers – Lucifer and DES. Understand asymmetric cryptographic systems and cryptology through*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 4 Understanding Systems Solve problems of Asymmetric key. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding Systems Solve problems of Asymmetric key*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition, ▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯.

Define or explain 8 Applying Cryptographic Systems Explain the concepts of Stream and block. (3 marks)

Answer: Begin with the standard definition or identity of *8 Applying Cryptographic Systems Explain the concepts of Stream and block*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition, ▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯.

Define or explain 6 Understanding ciphers Contents :Asymmetric (Public) Key Cryptographic Systems: Concept of PKCS, RSA Cryptosystem- Variants of RSA – Primality testing – Security of RSA – Merkle – Hellman – Security of Merkle – Hellman, ElGamal. Elliptical Curve Cryptography. Stream ciphers and block ciphers: The one time pad – Synchronous stream ciphers – Self-synchronizing stream ciphers – Feedback shift registers – Linear Complexity – Non-linear feedback shift registers – Stream ciphers based LFSRs. Explain the hash function as an application of cryptography with the. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding ciphers Contents :Asymmetric (Public) Key Cryptographic Systems: Concept of PKCS, RSA Cryptosystem- Variants of RSA – Primality testing – Security of RSA – Merkle – Hellman – Security of Merkle – Hellman, ElGamal. Elliptical Curve Cryptography. Stream ciphers and block ciphers: The one time pad – Synchronous stream ciphers – Self-synchronizing stream ciphers – Feedback shift registers – Linear Complexity – Non-linear feedback shift registers – Stream ciphers based LFSRs. Explain the hash function as an application of cryptography with the*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Illustrate the concepts of Digital signatures. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the concepts of Digital signatures*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain various secret sharing algorithms. (3 marks)

Answer: Begin with the standard definition or identity of *Explain various secret sharing algorithms*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Illustrate random number generators 3 Understanding Solve problems of digital signatures, secret. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate random number generators 3 Understanding Solve problems of digital signatures, secret*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain sharing algorithms and random number 5 Applying generators Contents : Digital Signatures: Properties, Generic signature schemes – Rabin - Lamport – Matyas Meyer, RSA – Multiple RSA and ElGamal Signatures – Digital signature standard – Blind Signatures- RSA Blind. Secret Sharing Algorithms: Threshold secret sharing – Shamir scheme, Blakley scheme and modular Scheme. Pseudo random number generators: Definition of randomness and pseudo-randomness – Statistical tests of randomness – Linear congruential generator – Modern PRNGs (a brief description). Describe various security threats to web applications/ servers and. (3 marks)

Answer: Begin with the standard definition or identity of *sharing algorithms and random number 5 Applying generators Contents : Digital Signatures: Properties, Generic signature schemes – Rabin - Lamport – Matyas Meyer, RSA – Multiple RSA and ElGamal Signatures – Digital signature standard – Blind Signatures- RSA Blind. Secret Sharing Algorithms: Threshold secret sharing – Shamir scheme, Blakley scheme and modular Scheme. Pseudo random number generators: Definition of randomness and pseudo-randomness – Statistical tests of randomness – Linear congruential generator – Modern PRNGs (a brief description). Describe various security threats to web applications/ servers and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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Module 4

Define or explain 4 Understanding defense mechanisms. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding defense mechanisms*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps,

□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Illustrate the mechanism of transmitting data 2 Understanding Explain vulnerabilities of access control and. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the mechanism of transmitting data 2 Understanding Explain vulnerabilities of access control and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain 4 Understanding securing mechanisms. Contents :Web application security- Key Problem factors – Core defense mechanisms- Handling user access- handling user input- Handling attackers – web spidering – Discovering hidden content. Transmitting data via the client – Hidden form fields – HTTP cookies – URL parameters – Handling client-side data securely – Attacking authentication – design flaws in authentication mechanisms – securing authentication Attacking access controls – Common vulnerabilities – Securing access controls Text / Reference T/R Book Title/Author Cryptography and Network Security: Principles and Practice, Global Edition, T1 7/E, William Stallings, Pearson Padmanabhan T R, Shyamala C and Harini N, “Cryptography and Security”, T2 Wiley Publications 2011. DafyddStuttard, Marcus Pinto, The Web Application Hacker’s Handbook, 2nd T3 Edition, Wiley Publishing, Inc. Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com. (3 marks)

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DafyddStuttard, Marcus Pinto, The Web Application Hacker’s Handbook, 2nd T3 Edition, Wiley Publishing, Inc. Online Resources Sl.No Website Link 1 https://www.tutorialspoint.com. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

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PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain basics of cryptography 2 Understanding Illustrate Symmetric Key Cryptographic**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **4 Understanding Systems Solve problems of Asymmetric key**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Illustrate the concepts of Digital signatures**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **4 Understanding defense mechanisms**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://www.tutorialspoint.com

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